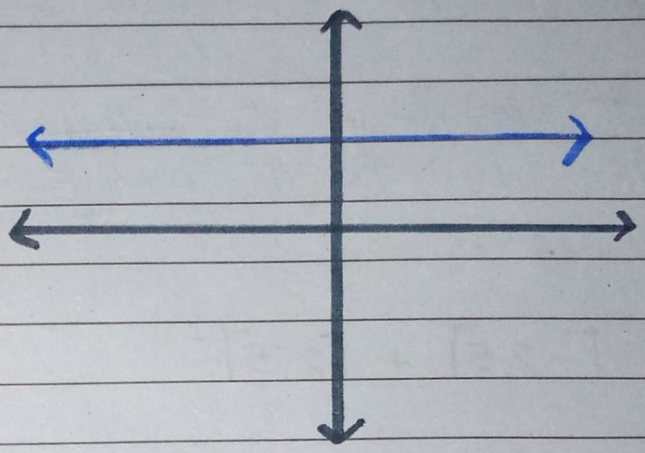
Basic Graph Shapes Of Common Functions :-

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These are the 8 essentials graphs you must memorize and understand perfectly. Each one builds your intuition for bigger topics.

1) Constant Function :-

$$f(x) = c$$



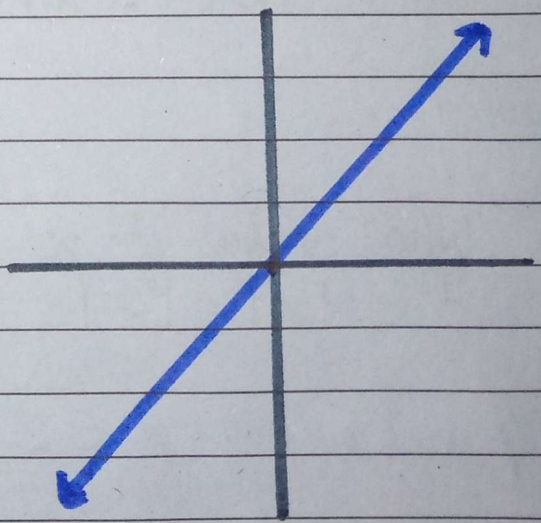
Where c is a constant number.

Shape : A horizontal line.

Graph Info : Domain : $(-\infty, \infty)$
Range : (c)

2) Indentity Function :-

$$f(x) = x$$



Shape : A diagonal line passing through the origin at 45°.

Domain : $(-\infty, \infty)$
Range : $(-\infty, \infty)$
Slope = 1
passes through origin $(0, 0)$

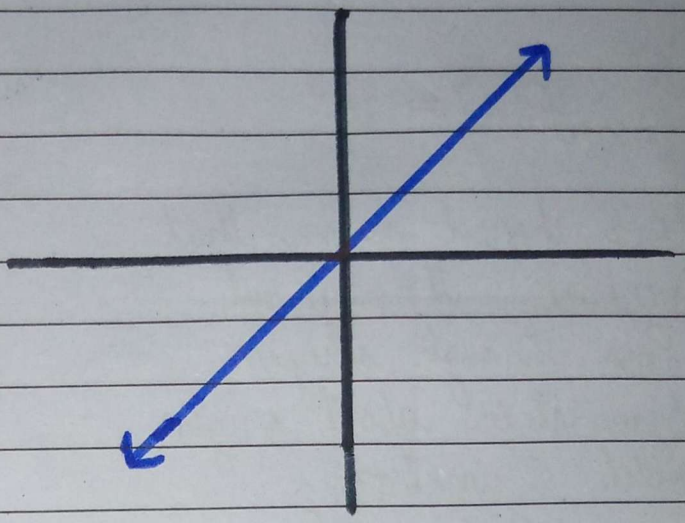
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3) Linear Function :-

$$f(x) = mx + b$$

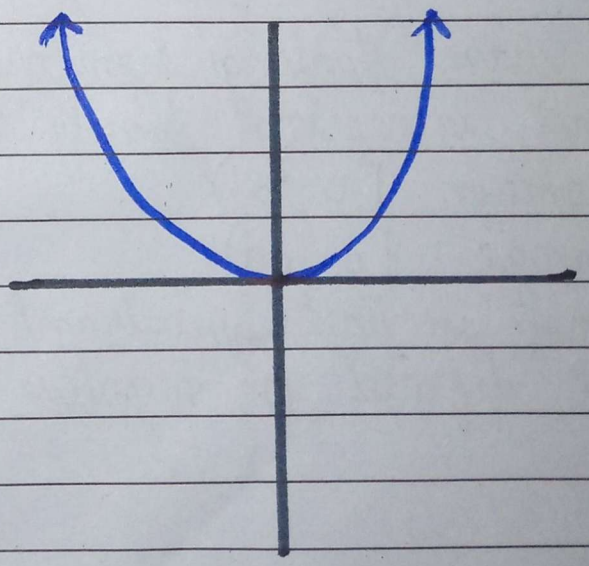


- Straight lines
- Constant slopes
- Domain & Range: $(-\infty, \infty)$
- Increasing if $m > 0$, decreasing if $m < 0$.

4) Quadratic Function :-

$$f(x) = x^2$$

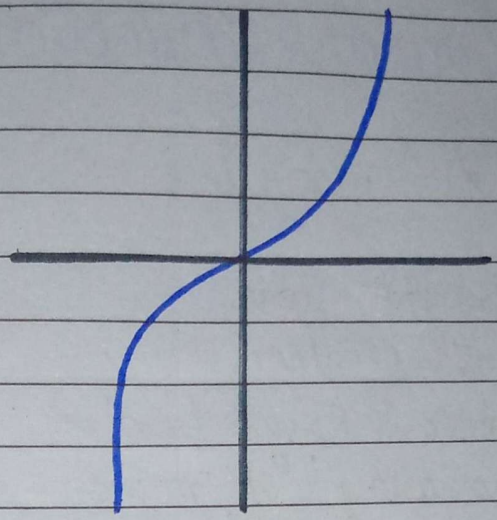
- A U-shaped curve called a parabola.
- Vertex : $(0, 0)$
- Symmetric about y-axis means (Even Function)
- Domain: $(-\infty, \infty)$, Range : $[0, \infty)$
- Changing slope



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5) Cubic Function :-

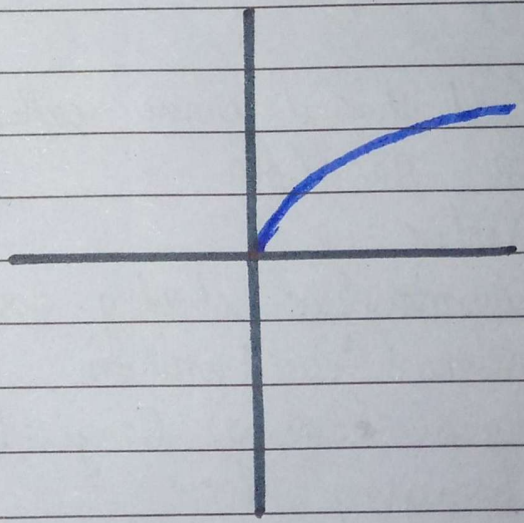
$$f(x) = x^3$$



- An S shaped curve that increase throughout.
- Pass through origin
- Symmetric about origin (odd Function)
- Domain & Range : $(-\infty, \infty)$
- Increasing everywhere

6) Square Root Function :-

$$f(x) = \sqrt{x}$$



- A curve starting from origin and increasing slowly.
- Domain : $[0, \infty)$
- Range : $[0, \infty)$
- Only in first quadrant.
- Not defined for negative values of x.

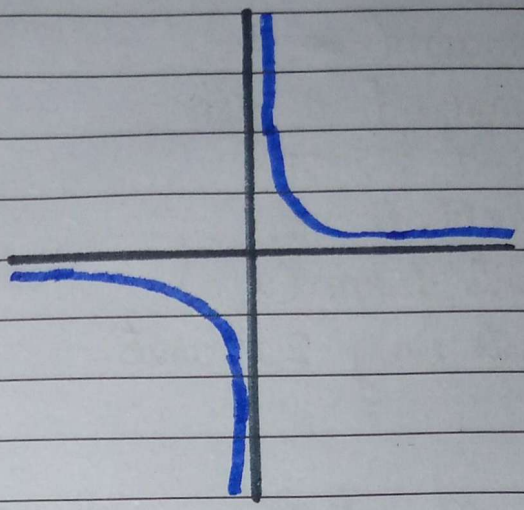
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7) Reciprocal Function :-

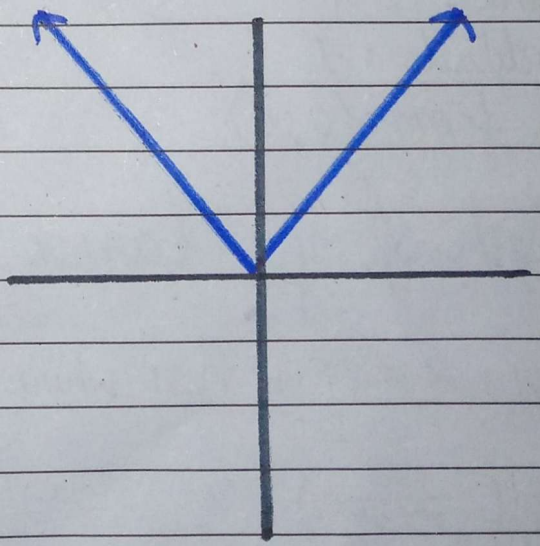
$$f(x) = \frac{1}{x}$$



- Two branches in 1st and 3rd quadrants.
- Hyperbola Shape
- Domain : $(-\infty, 0) \cup (0, \infty)$
- Range : $(-\infty, 0) \cup (0, \infty)$
- Asymptotes : $x = 0, y = 0$
- Never touches x or y axis.

8) Absolute Value Function :-

$$f(x) = |x|$$



- A V-shaped graph with vertex at origin.
- Domain : $(-\infty, \infty)$
- Range : $[0, \infty)$
- Symmetric about y-axis.
- Minimum at $x = 0$

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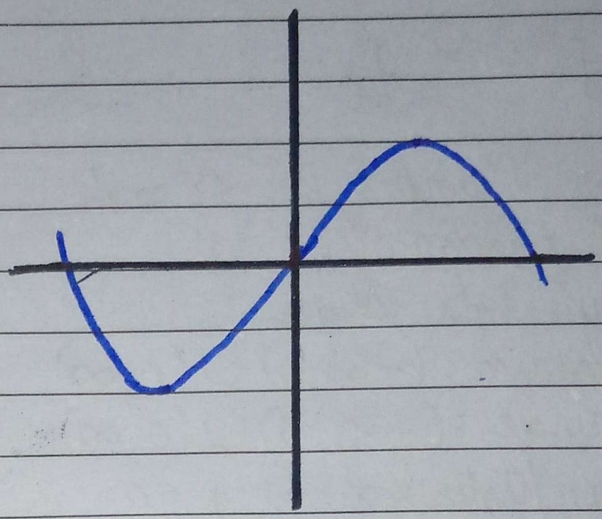
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Trignometric Graphs :-

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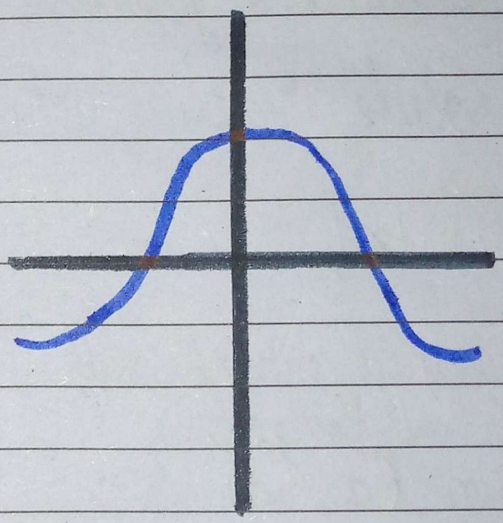
1) Graph of $y = \sin x$

- Domain: $(-\infty, \infty)$
- Range : $[-1, 1]$
- Period : 2π
- Amplitude : 1
- Starts from: $(0, 0)$
- Repeats every 2π units.



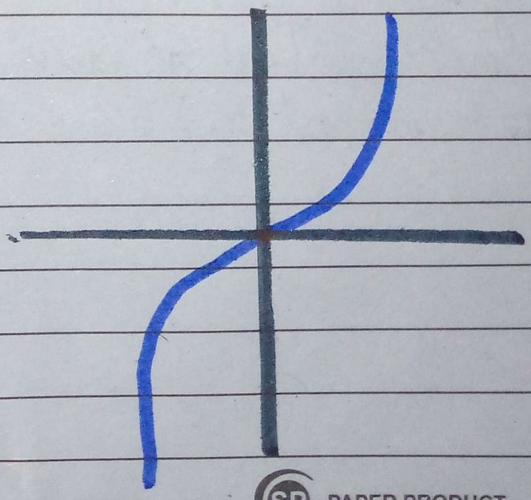
2) Graph of $y = \cos x$

- Domain : $(-\infty, \infty)$
- Range : $[-1, 1]$
- Period : 2π
- Amplitude : 1
- Starts from : $(0, 1)$



3) Graph of $y = \tan x$

- Domain : $x \neq \frac{\pi}{2} + n\pi$, where $n \in \mathbb{Z}$.
- Range : $(-\infty, \infty)$
- Period : π
- No amplitude (it goes to ∞)
- Steep S-curve shape.
- Has vertical asymptotes where cosine = 0.



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